

EXPERIMENT - 8

Date: 2022-10-21

TO DETERMINE THE REFRACTIVE INDEX OF THE MATERIAL OF PRISM BY I-D CURVE METHOD

APPARATUS REQUIRED:

- Glass Prism
- A sheet of paper
- Drawing board
- Fixing pins
- Scale protractor

THEORY:

If A is the angle of prism δ_m is the angle of minimum deviation of the prism, then the refractive index (μ) of the material of the prism is given by

$$\mu = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

OBSERVATION:

Least count of protector = 1°

OBSERVATION TABLE:

No. of Obs.	Angles of incidence (i)	Angles of deviation (δ)
1	35°	45°
2	40°	44°
3	45°	38°
4	50°	40°
5	55°	42°
6	60°	44°

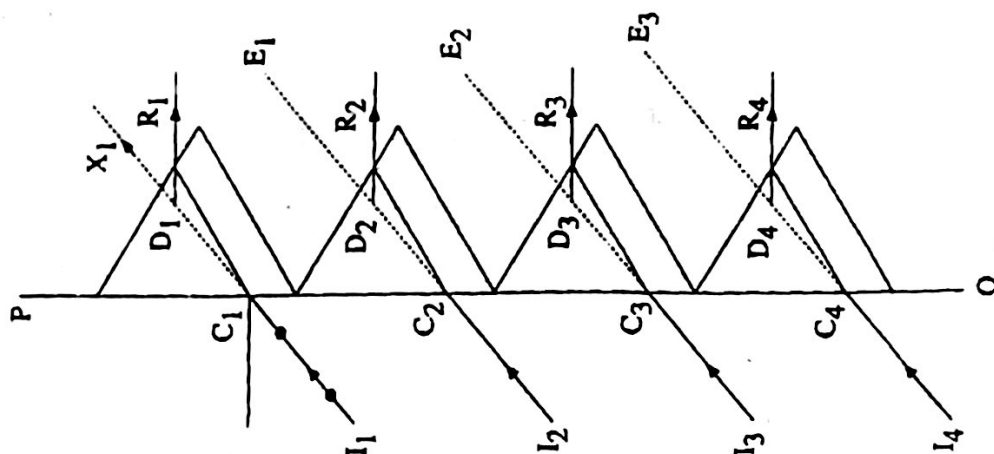


Fig.: Determination of angle of minimum deviation (δ_m)

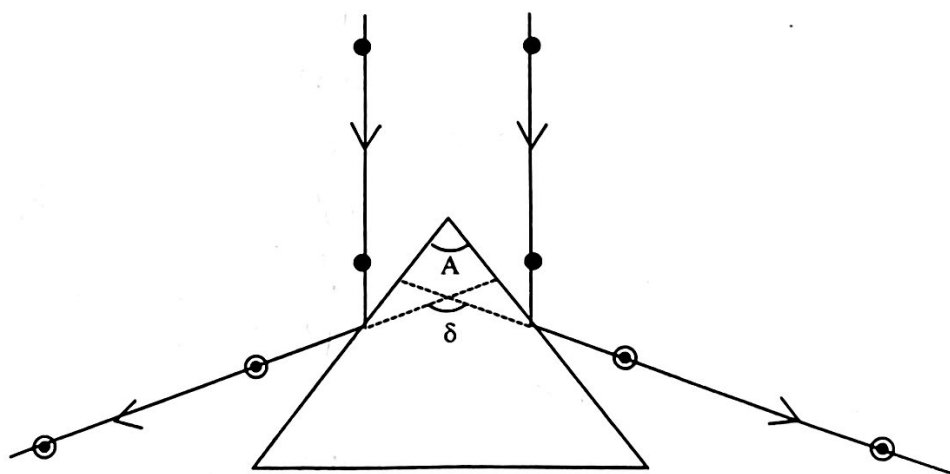
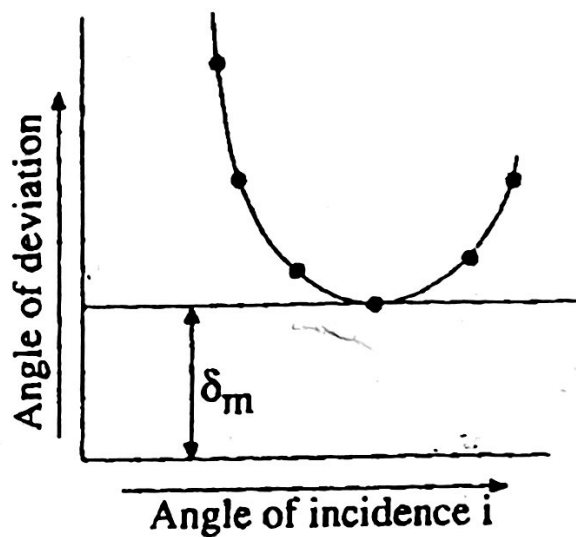


Fig.: Determination of Angle of Prism (A)

From graph, the angle of minimum deviation (δ_m) = 38°

CALCULATION:

The angle of prism (A) = 60°

The angle of minimum deviation from graph of I - D curve, (δ_m) = 38°

Now,

$$\mu = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)} = \frac{\sin\left(\frac{60 + 38}{2}\right)}{\sin\left(\frac{60}{2}\right)}$$

$$\therefore \mu = 1.5$$

RESULT:

Hence, the variation of angle of deviation with angle of incidence was studied and the refractive index of material of prism was calculated

PERCENTAGE ERROR:

Standard value of relative index (S.V.) = 1.5 (for glass)

$$\begin{aligned} \% \text{ Error} &= \left| \frac{\text{S.V.} - \text{O.V.}}{\text{S.V.}} \right| \times 100\% \\ &= \left| \frac{1.5 - 1.5}{1.5} \right| \times 100\% \\ &= 0\% \end{aligned}$$

CONCLUSION:

Hence, from the above experiment, we studied the variation of angle of deviation with angle of incidence & calculated the 'n' of material of prism with 0% error.

SOURCES OF ERROR:

- 1) Improper measurement of angle.
- 2) Two image pins may be too near to each other.
- 3) The object pins and image pins may not lie in same straight line while viewing in prism.

PRECAUTIONS:

- 1) The separation between the points should be about 10cm.
- 2) Arrow head should be used to denote the direction of rays.
- 3) The prism should not be moved from its original place.
- 4) One eye should be closed while observing the image.

VIVA QUESTIONS:

1. What do you mean by I-D curve?
2. Define refractive index of a material in terms of the speed of light.
3. What is meant by the statement, μ of glass is 1.5?
4. What is the value of critical angle of a material whose μ is 2.4?
5. What are the conditions for minimum deviation in a prism?
6. What is meant by grazing incidence and grazing emergence?
7. How does angle of deviation vary with angle of incidence?

1. The curve which shows the variation of angle of deviation and angle of incidence on a prism is called I-D curve.

2. Refractive index is the ratio of speed of light in vacuum to the speed of light in medium.

3. It means that the direction of refractive ray deviate towards the normal by two third of the direction of original ray. i.e. incident ray.

4. We know,

$$\mu = \frac{1}{\sin c}$$

$$\text{or, } 2.4 = \frac{1}{\sin c}$$

$$\text{or, } \sin c = \frac{1}{2.4}$$

$$\text{or, } c = \sin^{-1} \left(\frac{1}{2.4} \right)$$

$$\text{or, } c = 24.62^\circ$$

$$\therefore c = 24.62^\circ$$

5. The condition for minimum deviation in a prism is $i = e$ and $r_1 = r_2$.

6. When a ray of light is incident on a prism with angle of incidence 90° , such incidence is called grazing incidence. When a ray of light is emergent from a prism with angle of emergence is called grazing emergence.

7. When the angle of incidence increases, angle of deviation first decreases, becomes minimum and then increases.

